

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	42.0 / Female
Department	Gynaecology
Diagnosis	Acute cystitis
UTI Classification	Uncomplicated UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Dysuria;Lower abdominal pain

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	37.0	%	20 - 40
Wbc	9200.0	cells/ μ L	4000 - 11000
Polymorphs	65.0	%	40 - 75
Crp	12.0	mg/L	< 10
Rft Serum Creatinine	0.8	mg/dL	0.6 - 1.3
Serum Uric Acid	4.2	mg/dL	3.5 - 7.2
Blood Urea	28.0	mg/dL	7 - 20
Cue Pus Cells	14.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	2.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Candida albicans
Bacteria Type	Fungus (Yeast; the most common cause of candiduria)

Antibiotic Susceptibility Profile

Predicted Sensitive	Amphotericin B, Voriconazole
Predicted Resistant	Fluconazole

Recommended Therapy

Voriconazole

Dosage: 200 mg orally every 12 hours for 7-14 days (standard adult dose).

Precautions: Check baseline liver function tests; use with caution in hepatic impairment. Monitor for visual disturbances and photosensitivity. Review concomitant medications for CYP2C19/CYP3A4 interactions. No dose adjustment needed for normal renal function (creatinine 0.8 mg/dL). Avoid in pregnancy.

Rationale: Voriconazole is a broad-spectrum azole active against Candida albicans and is listed as a predicted sensitive agent. It can be given orally, which is convenient for an uncomplicated cystitis case, and avoids the nephrotoxicity associated with amphotericin B.

Amphotericin B (liposomal formulation)

Dosage: 3 mg/kg IV once daily (liposomal amphotericin B) for 7-14 days, adjusted to clinical response.

Precautions: Monitor renal function (serum creatinine, electrolytes) closely despite current normal creatinine (0.8 mg/dL). Pre-hydrate patient to reduce nephrotoxicity. Watch for infusion-related reactions (fever, chills) and electrolyte disturbances (hypokalemia, hypomagnesemia). Use with caution in patients with pre-existing renal disease.

Rationale: Amphotericin B is a potent fungicidal agent with activity against Candida albicans and is a predicted sensitive option. It is reserved for cases where azole therapy is contraindicated or ineffective, providing an alternative for severe or refractory infection.

Antibiotic Drug History

Clinical Summary

Infection: Acute uncomplicated cystitis (bladder infection) in a 42-year-old female.

Predicted pathogen: *Candida albicans* (fungal cystitis).

Resistance profile:

- Resistant to fluconazole (predicted).

Sensitivity profile (predicted):

- Amphotericin B (including liposomal formulation) - fungicidal.
- Voriconazole - broad-spectrum azole active against *C. albicans*.

Recommended therapy

1. **Voriconazole** 200 mg PO q12 h for 7-14 days.

Rationale: Oral route is convenient for an uncomplicated case and provides effective azole coverage without the nephrotoxicity of amphotericin B.

Precautions:

- Obtain baseline liver function tests; use cautiously in hepatic impairment.
- Monitor for visual disturbances, photosensitivity, and drug-drug interactions (CYP2C19/3A4).
- No renal dose adjustment needed (creatinine 0.8 mg/dL).
- Contraindicated in pregnancy.

2. **Liposomal Amphotericin B** 3 mg/kg IV daily for 7-14 days (adjust per response).

Rationale: Potent fungicidal agent reserved for cases where azole therapy is unsuitable or fails.

Precautions:

- Pre-hydrate and monitor renal function, electrolytes (K⁺, Mg²⁺) despite current normal creatinine.
- Watch for infusion-related reactions (fever, chills).
- Use with caution in pre-existing renal disease.

Key considerations:

- Ensure the patient is not pregnant before initiating voriconazole.
- Baseline LFTs and periodic monitoring are essential during azole therapy.
- If renal function deteriorates or azole contraindications arise, switch to liposomal amphotericin B with appropriate supportive measures.